

SURESH LAMA

Electronics Specialist KONE R&D | Hyvinkää, Finland | +358403594439 || [LinkedIn](#)

SUMMARY

Electronics Specialist with 7+ years of experience in PCBA development and product change management within KONE R&D. Proven track record of leading 16+ CRs resolving semiconductor obsolescence strengthening business continuity, improving quality, bringing cost saving and contributed to D200 to achieve D4 milestones as second CDE.

EDUCATION

Metropolia UAS, Helsinki, Finland

2013-2017

- Bachelor of Electronics Engineering

TECHNICAL SKILLS

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- PCB and PCBA manufacturing with DFM/DFT, Schematic/Layout (CR8000), Alternative component and semiconductor based on KONE internal design guidelines, Knowledge of EMC (Emission & immunity)
 - Component and PCBA validation engineering, Digital and Analog electronics, Embedded C programming,
 - Production testing tooling ICT, FCT and FMT,
 - Probability of failure calculation PFH avg,
 - ProgeCAD for 2D drawing,
 - CI CD, Docker, Cmake, Embedded C, RTOS basics, STM32Cube IDE and VsCode IDE

KONE TECHNICAL SKILLS

KCE system, Door DFSI protocol, familiar with LON & TTS communication, PESSERAL system, PPMT, Order form, CR, CA, KPPAP, Jira, Jama,

DFR process. Reuse check, CTQ process check, Technology and Reliability plan, HW V&V plan, DFMEA

WORK EXPERIENCE

KONE R&D,

Electronics Specialist, Hyvinkää, Finland

2022-till

Doors Innovation and value Engineering,

Car & Door tribe

- As second CDE for D200 project, facilitated completion of checklists of technology and reliability plan and ensured all hardware verification and validation activities were finished to achieve D4 milestone. Proposed the HW solution to address the PCBA failure from in-house piloting and proposed design update to prevent conducted emission failure in pre-compliance EMC testing, led the successful update of D200 door operator drawings from outsourced engineering partner and coordinated first 10 successful customer piloting in Finland.
- Responsible for elevator door drive, COL and its electronics PCM, Successful completion of more than 15 Change requests addressing scope of CR that includes cost saving, feedback from frontlines, supplier PCBA production, quality and reliability following KPPAP with local SQM.
- Introduction of second supplier for bare PCB, PCBA, PMSM motor, Transformers, cable assembly and electronic components bringing 20% cost reduction of door operator price and de-risking single supplier's problem.
- Successfully redesigned and validated D15& D2 encoder PCBA to avoid component obsolescence in time and reduced business continuity risk.
- Made root cause failure analysis of D10 FCT production testing failure and proposed solution that involves test SW update instead of actual tester to prevent production delay, validated and successfully implemented.
- Successfully led cost saving (20 Euro/elevator) CR for the development of AMD10REL PCBA + enclosure from POC- prototyping till CR3. Reviewed and provided technical support for AMD10REL door operator drawings update to design team. MVP (HW+SW+mechanics) produced and tested with cross functional team collaboration with KCECCG door SW team and mechanical team.

KONE R&D,

Senior Engineer, Hyvinkää, Finland

2021-2022

EU PCBA engineering team,

- During global semiconductor crisis in the Covid period, more than 20 second sources introduced to semiconductors including door drive MCU by collaborating with supplier, SQM and component engineering team and prevented business continuity risk and prevented cost increase in buying on shelf component.
- Successful Proof of Concept validation for new KCEPIT inspection to use gyroscope-based detection [KONE patent] instead of reed switch needed for business case of cost saving and potential patent hassle from current supplier. Schematic design of KCEPIT inspection PCBA and later led development of [electronics, mechanics, and cabling] for cost saving and patent avoidance from existing supplier ensuring design.
- Schematic design and layout of universal dongle design as part of predictive maintenance solution for automatic building doors. Dongle could connect 3rd party door operator with CAN, RS232, RS485 or ethernet interface.
- Worked as core team member from day1 to create template and guidelines for probability of failure calculation acc. to IEC61508 and rule to calculate PFD/PFH for KCE PESSERAL boards.
- Successfully conducted the verification and validation of COP's UIDDUC PCBA including FCT tester readiness, functional testing with KCE simulator and EMC till released for mass production.
- Functional testing of KCEMSC and KCECCG board's components.
- R&D modular prototypes testing and programming fixture (HW+SW) development that program Proto boards, test basic function, set parameters and store test results locally. Use of STM32 CubeIDE and STM32L4Q5Gt6 based MCU for running fixture and python script.

KONE NRD,

PCBA specialist, NRD, Netherland

2019-2020

Line manager Harri Anttila

- Three successful PCBA NPI made with KONE design guidelines where ICT, KONE FCT and KONE FMT testers were first implemented and SEB handover and two new vendors selected,
- Redesign of new generation automatic building door drive, Universal program Switch and 24/7 predictive maintenance solution.

KONE (Adecco), Design Engineer, Hyvinkää, Finland

2017-2021

Validation engineering includes test planning and setups, functional testing for prototypes and Spice simulation.

- Implementation of python test Script and python based UI to automate the programming of proto CPUc boards and parameter setting that saves time and prevent manual human error.
- Functional testing of KCECCG, KCEMSC, KCECPUc KNSMUL, KNSKIB and battery Monitoring feature test.

PROFESSIONAL DEVELOPMENT TRAINING

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| • Design for Six sigma Yellow Belt, | 2023 |
| • Functional Safety IEC61508 | 2020 |
| • SAFe Scrum Master | 2022 |

LANGUAGE

English - Fluent	Nepali - Fluent
Finnish - Intermediate	Hindi - Fluent